

Examination Statistical Methods

Linköpings Universitet, IDA, Statistik

Course code and name:	732A83 Statistical Methods
Date:	2025/02/25, 8–12
Examinator:	Krzysztof Bartoszek
Allowed aids:	Pocket calculator Table with common formulæ and moment generating functions (distributed with the exam) Table of integrals (distributed with the exam) Tables with distributions, probabilities and percentage points (distributed with the exam) One double sided A4 page with own hand written notes
Grades:	A= $[18 - \infty)$ points B= $[16 - 18)$ points C= $[14 - 16)$ points D= $[12 - 14)$ points E= $[10 - 12)$ points F= $[0 - 10)$ points
Instructions:	Write clear and concise answers to the questions. Reporting on the exam a probability outside the $[0, 1]$ interval or a negative variance without a comment that a mistake must have been made somewhere will result in a NEGATIVE point for each such instance.

Problem 1 (2p)

A telemarketer chooses numbers of prospective clients using a random number generator, that generates each of the 7 digits of the telephone number independently and uniformly on $\{0, \dots, 9\}$, with the exception that the first digit cannot be 0.

1. (1p) Calculate the probability that the telephone number will start and end with the digit 5.
2. (1p) What is the probability that two consecutively sampled telephone numbers will start with the same two digits?

Problem 2 (2p)

Consider two events A and B such that $P(A) = 0.3$, $P(A|B) = 0.2$ and $P(B|A) = 0.4$

1. (1p) What is the probability that event B takes place?
2. (1p) What is the probability that neither event A nor B takes place?

Problem 3 (3p)

Consider a discrete random variable $X \in \{0, 1, 2, 3, 4\}$ with probability distribution $P(\{0\}) = 0.4$, $P(\{1\}) = 0.1x$, $P(\{2\}) = 0.05x$, $P(\{3\}) = 0.05x$, $P(\{4\}) = 0.1x$.

1. (1p) Calculate x .
2. (1p) Let $F_X(\cdot)$ be the cumulative distribution function of X . Find $F_X(-1)$, $F_X(2)$ and $F_X(10)$.
3. (1p) Find the expectation and variance of X .

Problem 4 (2p)

A certain production plant produces cogwheels. Production is imperfect and their diameter is subject to an error, X that has density function $f_X(x) = c(x+2)(x-3)$ for $-2 < x < 3$.

1. (1p) Calculate c .
2. (1p) If a diameter deviates by more than 1, then the cogwheel has to be rejected. Calculate the percentage of rejected cogwheels.

Problem 5 (4p)

We observe the following i.i.d. sample 1.2, 4.5, 2.6, 0.4, 3.3, 3.0, 1.5, 3.5, 2 which is modelled by the density function

$$f(x) = \frac{x}{\theta^2} e^{-x/\theta} x \geq 0.$$

1. (2p) Find the maximum likelihood estimator of θ and calculate its value.
2. (2p) Find the method of moments estimator θ and calculate its value. Compare with the maximum likelihood one.

Problem 6 (4p)

Consider a stochastic process $\epsilon_n \in \{0, 1\}$ such that for each n $P(\epsilon_n = 0) = 1 - p$ and $P(\epsilon_n = 1) = p$. Then consider the sum $S_n = \epsilon_1 + \dots + \epsilon_n$.

1. (2p) What is the distribution of S_n ?
2. (2p) Calculate the mean and variance of S_n for $p = 0.5$.

Problem 7 (3p)

Consider a Poisson process, $N(t)$, with intensity 1.

1. (1p) What is the probability that $P(N(1) = 2)$?
2. (1p) What is the probability that $P(N(4) - N(2) = 1)$?
3. (1p) What is the expected time between two jumps of the process?